

CEERS | Barred Galaxy Candidate Report

CEERS mosaic and detection context

Mosaic: hlsp_ceers_jwst_nircam_nircam8_f444w_dr0.6_i2d.fits (F444W) | Pixel scale: 0.0300 arcsec/pixel

Diffraction-based PSF estimate: FWHM 0.141 arcsec (4.70 pixels); Gaussian sigma 1.99 pixels; PSF half-maximum disk area 17.32 pixel² (0.0156 arcsec²).

Candidate summary

Barred fraction: (4 strongly + 38 weakly) / 305 detected galaxies = 13.8%

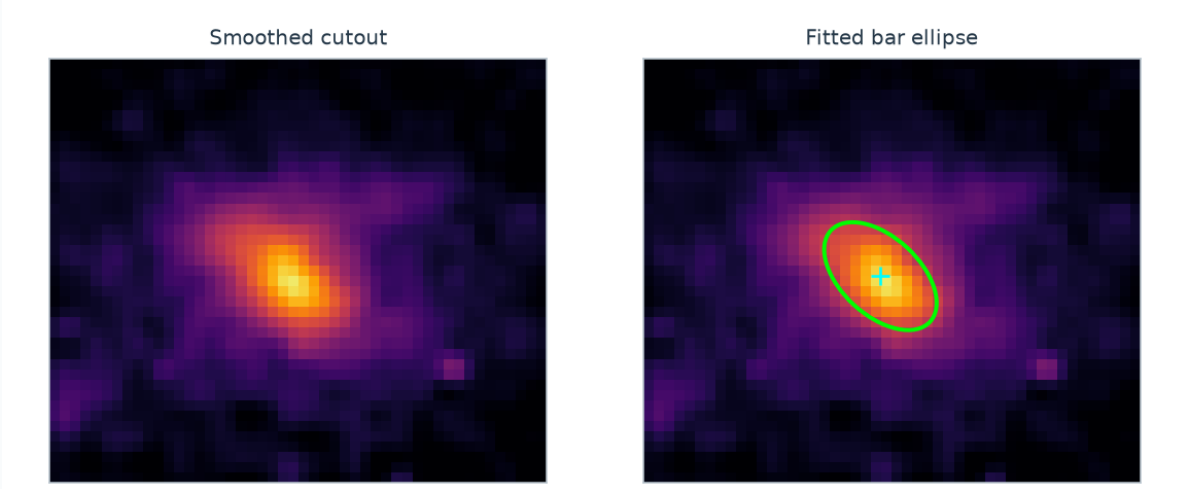
Strong candidates: 4 | Weak candidates: 38 | Total candidates: 42 | Median bar b/a: 0.545 | Median bar eccentricity: 0.838

Method: high-intensity connected candidate region, weighted second moments, and the notebook's b/a thresholds (strong < 0.40; weak 0.40-0.60). Bar PA follows the fitted major axis, measured east of north. The line shows the fitted moment ellipse and the cross its center; neither marks an independently fitted structural bar. Bar length is intentionally blank.

Weakly Barred Galaxies

CEERS | physical source 00016

Weakly Barred

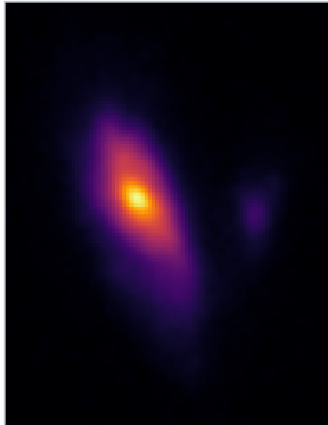


Bar b/a	0.580	Eccentricity	0.815
Bar PA	177.9 deg	Estimated bar length	_____
Galaxy sky center RA 215.044666 deg Dec +52.882250 deg			

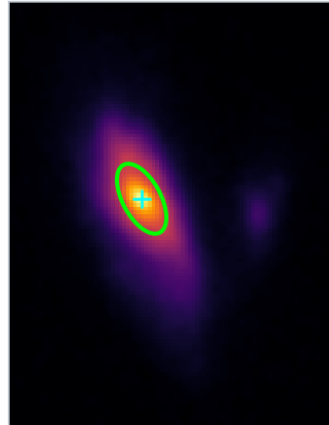
CEERS | physical source 00065

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.494

Eccentricity 0.869

Bar PA 158.9 deg

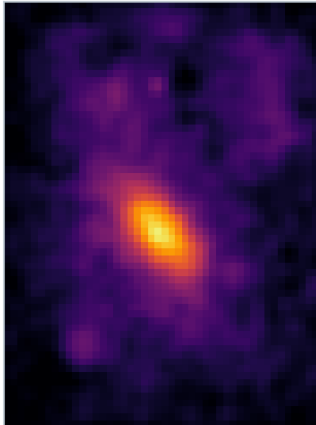
Estimated bar length _____

Galaxy sky center RA 215.020491 deg | Dec +52.867833 deg

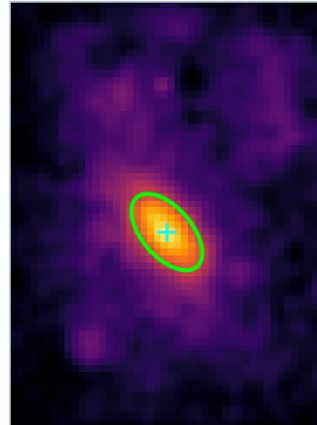
CEERS | physical source 00077

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.542

Eccentricity 0.840

Bar PA 171.7 deg

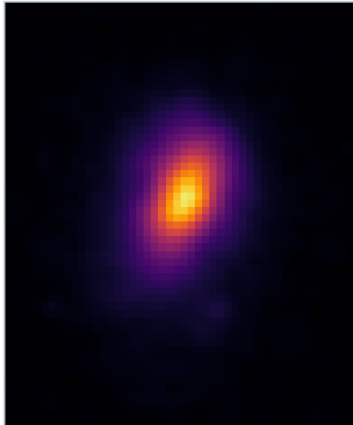
Estimated bar length _____

Galaxy sky center RA 215.077963 deg | Dec +52.908736 deg

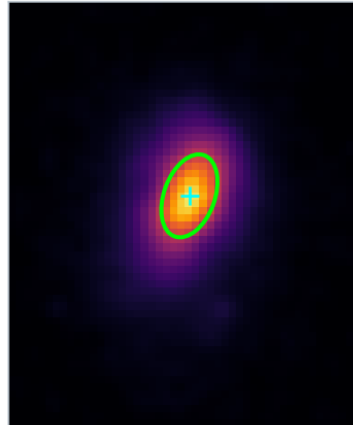
CEERS | physical source 00083

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.589

Eccentricity 0.808

Bar PA 110.3 deg

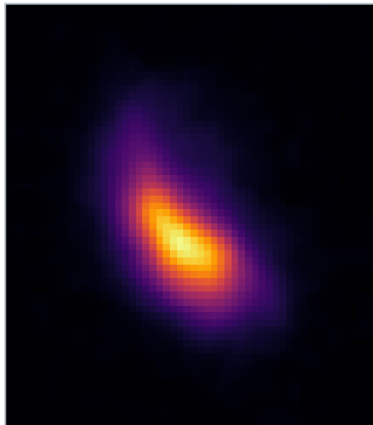
Estimated bar length _____

Galaxy sky center RA 215.047815 deg | Dec +52.887687 deg

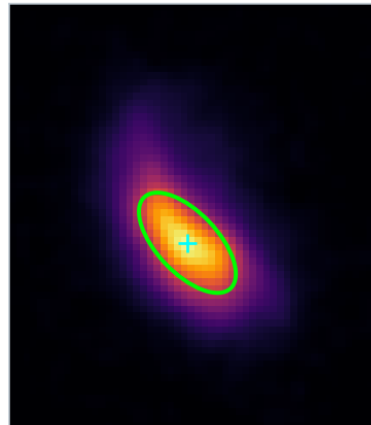
CEERS | physical source 00135

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.483

Eccentricity 0.876

Bar PA 174.0 deg

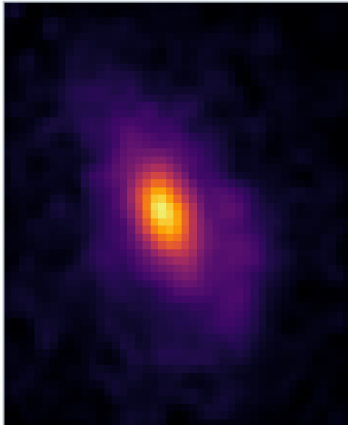
Estimated bar length _____

Galaxy sky center RA 215.062605 deg | Dec +52.899849 deg

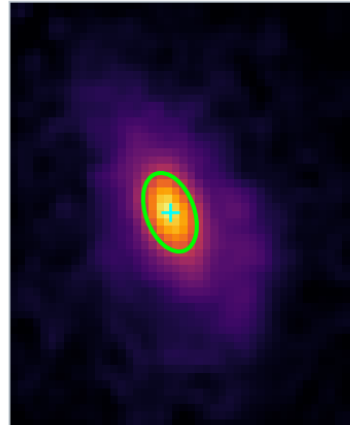
CEERS | physical source 00141

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.590

Eccentricity 0.807

Bar PA 151.1 deg

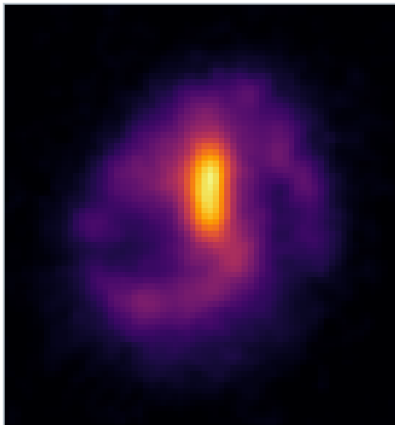
Estimated bar length _____

Galaxy sky center RA 215.026305 deg | Dec +52.874316 deg

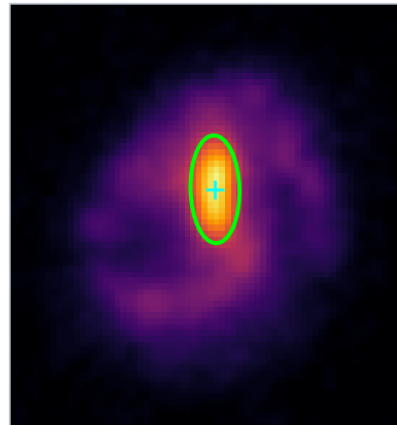
CEERS | physical source 00152

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.455

Eccentricity 0.890

Bar PA 132.0 deg

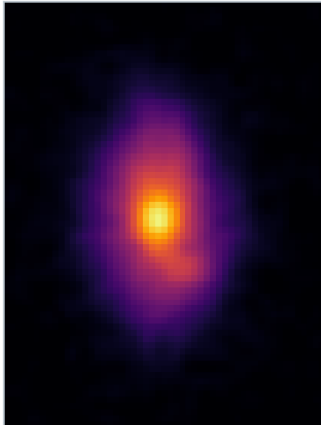
Estimated bar length _____

Galaxy sky center RA 215.072029 deg | Dec +52.907108 deg

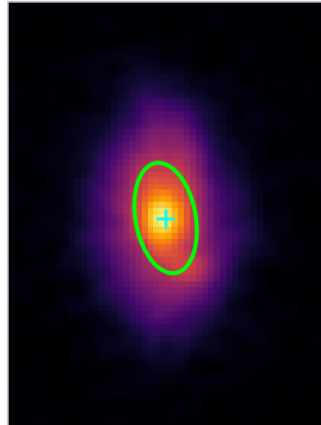
CEERS | physical source 00225

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.541

Eccentricity 0.841

Bar PA 140.9 deg

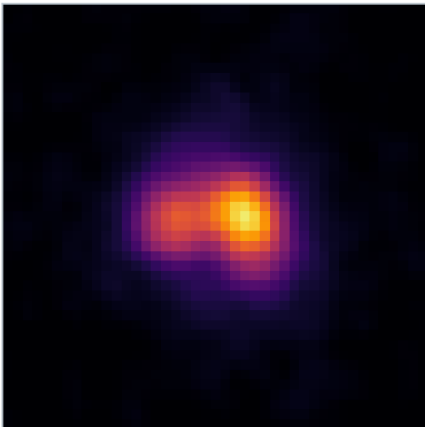
Estimated bar length _____

Galaxy sky center RA 215.013400 deg | Dec +52.867794 deg

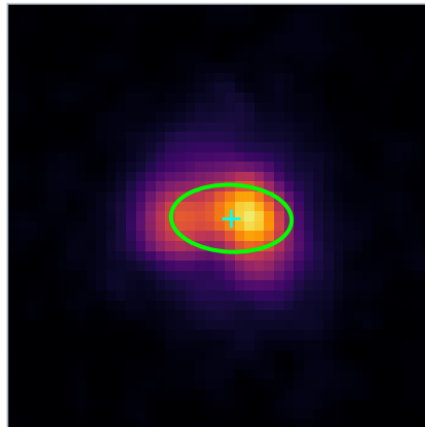
CEERS | physical source 00248

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.555

Eccentricity 0.832

Bar PA 38.3 deg

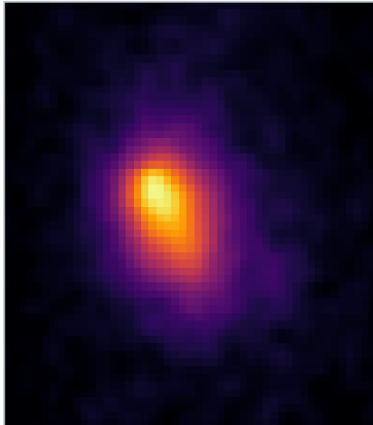
Estimated bar length _____

Galaxy sky center RA 215.062091 deg | Dec +52.902818 deg

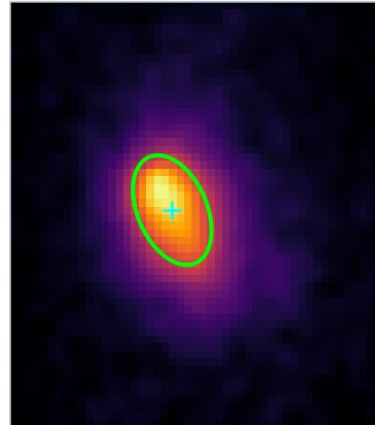
CEERS | physical source 00252

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.598

Eccentricity 0.801

Bar PA 154.2 deg

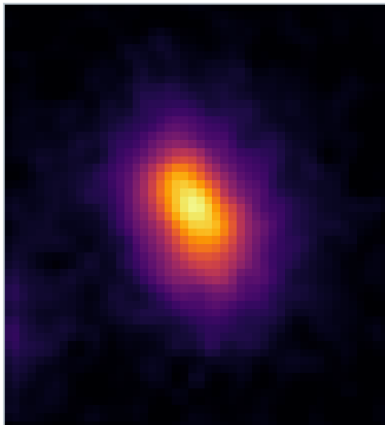
Estimated bar length _____

Galaxy sky center RA 214.987686 deg | Dec +52.850401 deg

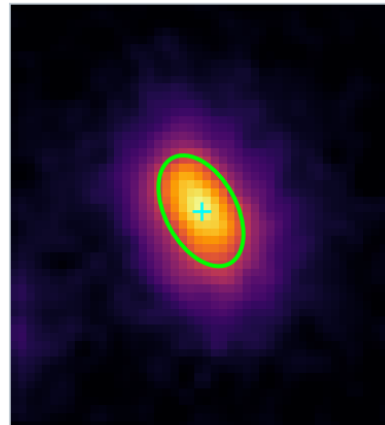
CEERS | physical source 00277

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.592

Eccentricity 0.806

Bar PA 159.2 deg

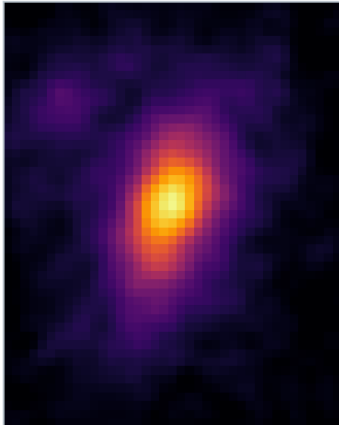
Estimated bar length _____

Galaxy sky center RA 214.999198 deg | Dec +52.859550 deg

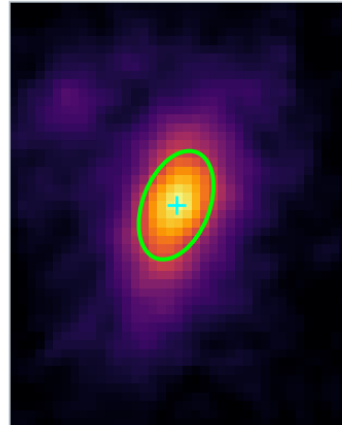
CEERS | physical source 00303

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.590

Eccentricity 0.807

Bar PA 109.0 deg

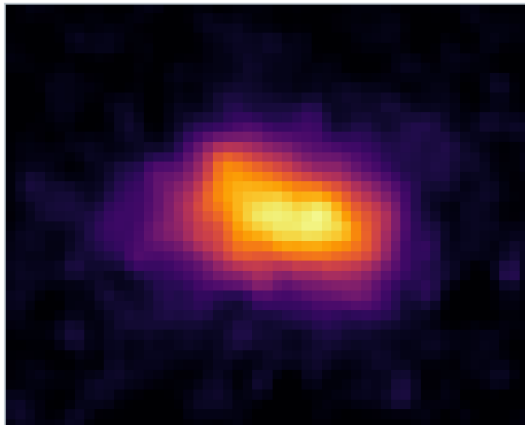
Estimated bar length _____

Galaxy sky center RA 215.064345 deg | Dec +52.906619 deg

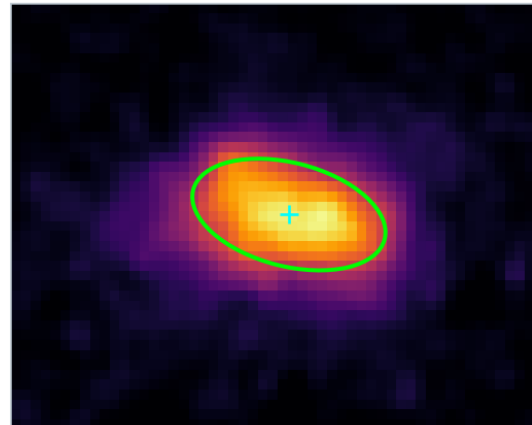
CEERS | physical source 00371

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.522

Eccentricity 0.853

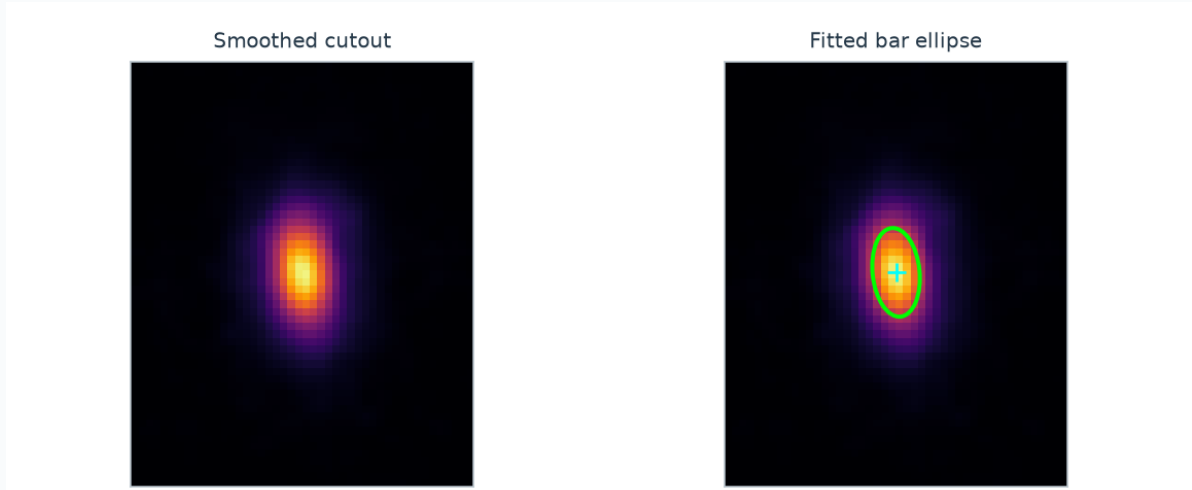
Bar PA 26.1 deg

Estimated bar length _____

Galaxy sky center RA 215.047542 deg | Dec +52.896543 deg

CEERS | physical source 00374

Weakly Barred



Bar b/a 0.529

Eccentricity 0.849

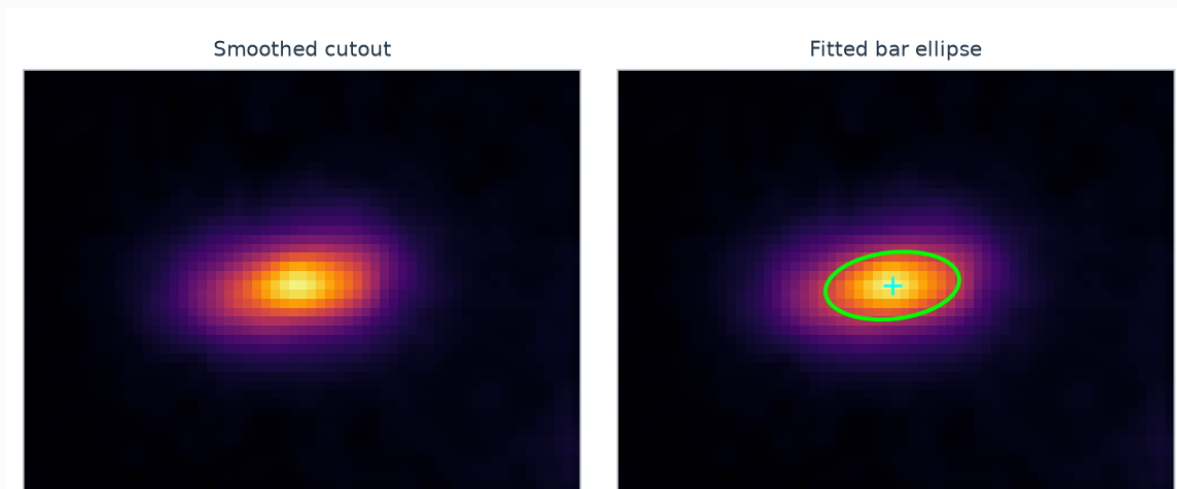
Bar PA 136.3 deg

Estimated bar length _____

Galaxy sky center RA 215.050211 deg | Dec +52.898590 deg

CEERS | physical source 00457

Weakly Barred



Bar b/a 0.494

Eccentricity 0.869

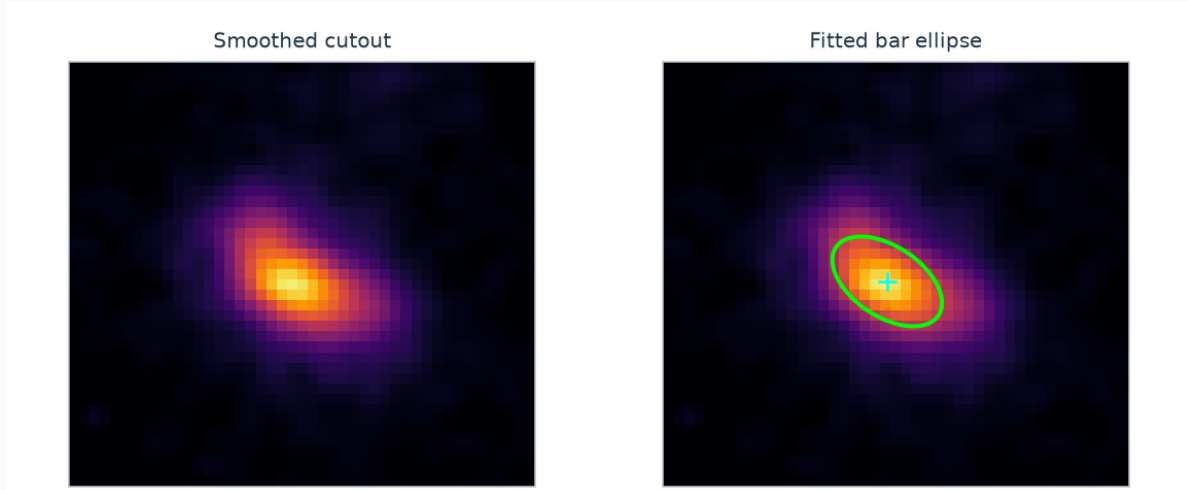
Bar PA 47.0 deg

Estimated bar length _____

Galaxy sky center RA 215.068512 deg | Dec +52.914339 deg

CEERS | physical source 00458

Weakly Barred



Bar b/a 0.570

Eccentricity 0.822

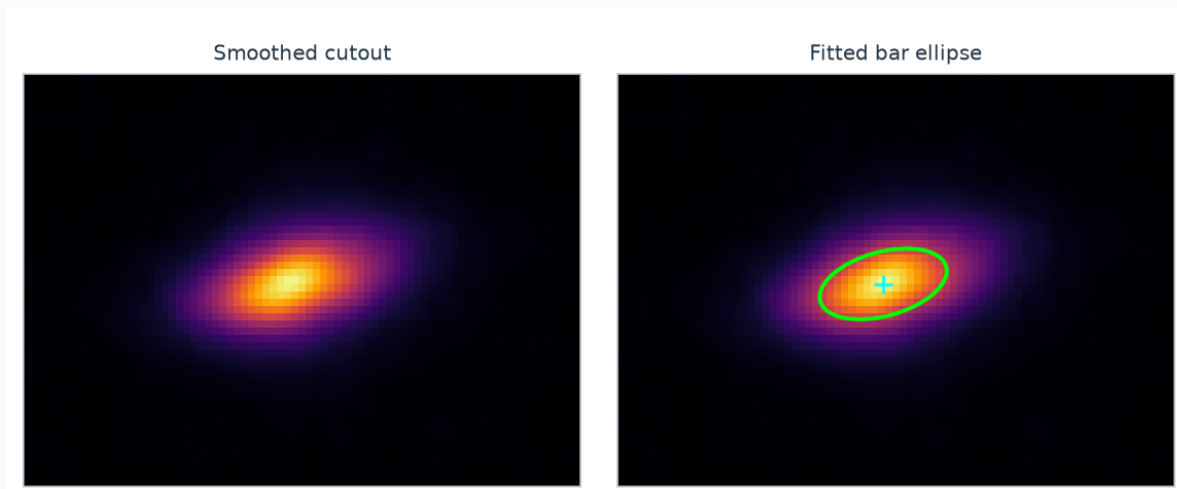
Bar PA 6.9 deg

Estimated bar length _____

Galaxy sky center RA 215.006354 deg | Dec +52.870363 deg

CEERS | physical source 00459

Weakly Barred



Bar b/a 0.489

Eccentricity 0.872

Bar PA 55.7 deg

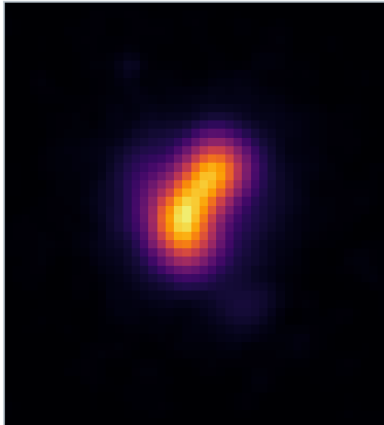
Estimated bar length _____

Galaxy sky center RA 214.985285 deg | Dec +52.855687 deg

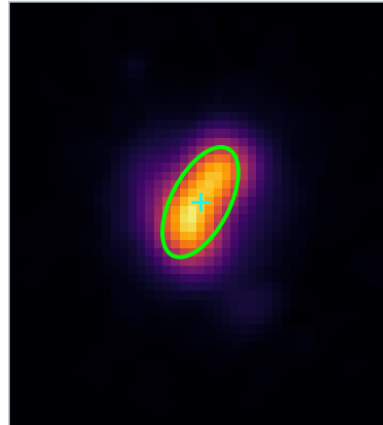
CEERS | physical source 00491

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.484

Eccentricity 0.875

Bar PA 102.9 deg

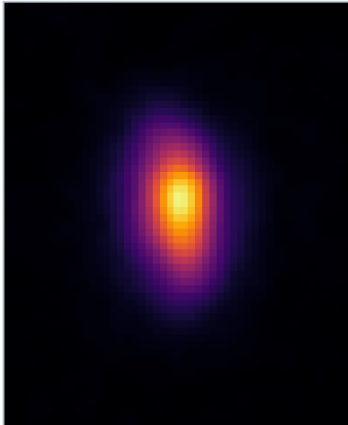
Estimated bar length _____

Galaxy sky center RA 214.993160 deg | Dec +52.862355 deg

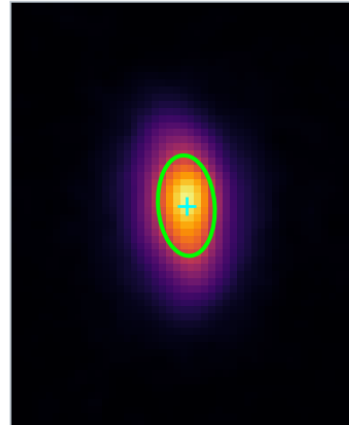
CEERS | physical source 00639

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.563

Eccentricity 0.827

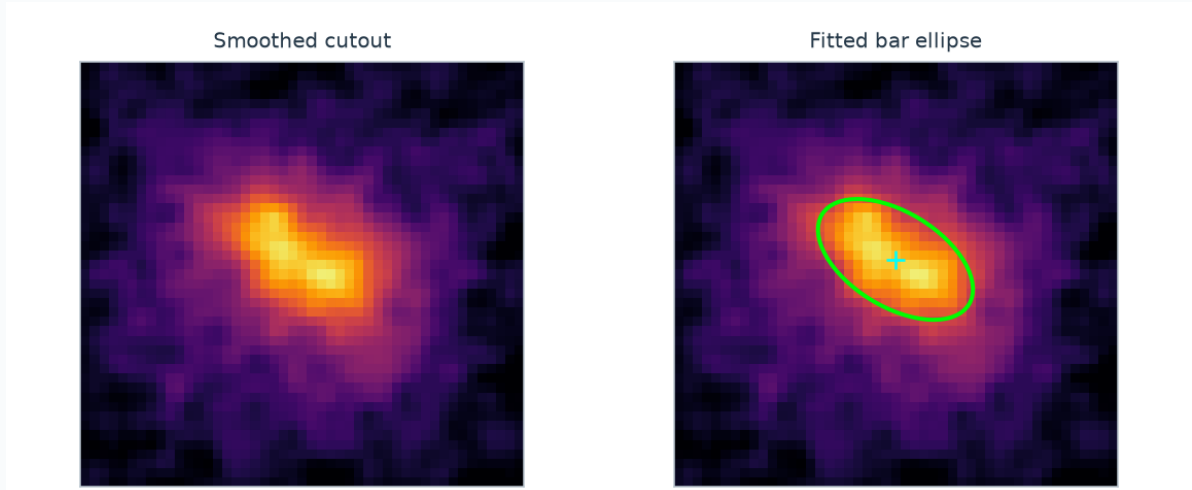
Bar PA 135.1 deg

Estimated bar length _____

Galaxy sky center RA 215.055083 deg | Dec +52.912097 deg

CEERS | physical source 00674

Weakly Barred



Bar b/a 0.548

Eccentricity 0.837

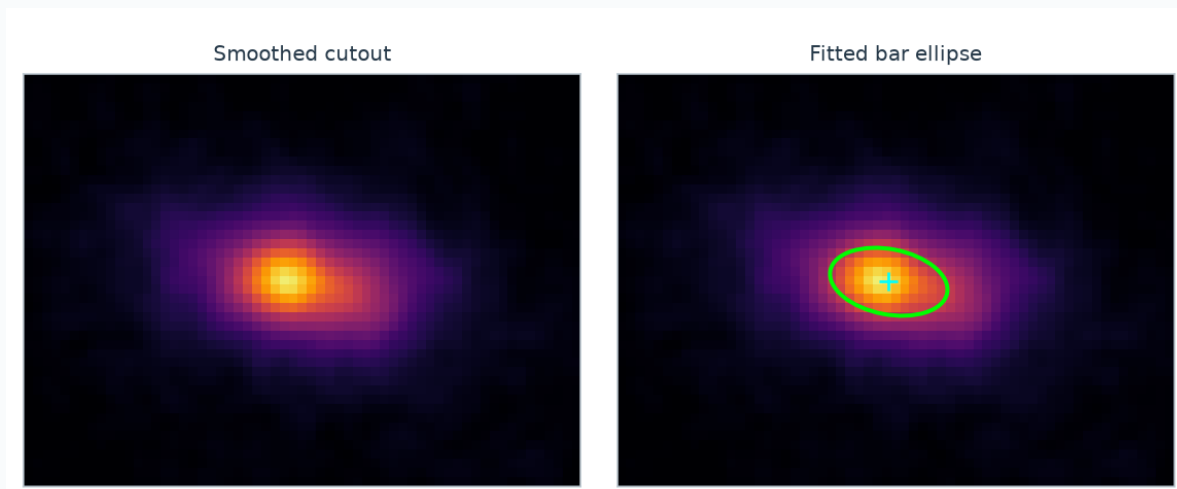
Bar PA 9.1 deg

Estimated bar length _____

Galaxy sky center RA 215.060155 deg | Dec +52.917252 deg

CEERS | physical source 00778

Weakly Barred



Bar b/a 0.548

Eccentricity 0.836

Bar PA 29.5 deg

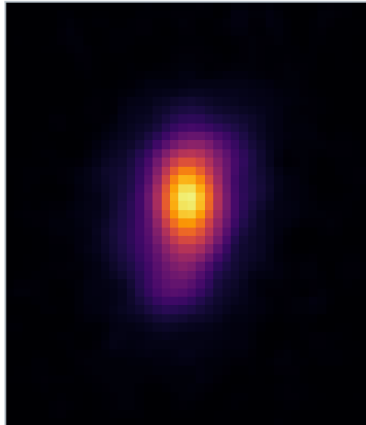
Estimated bar length _____

Galaxy sky center RA 214.974785 deg | Dec +52.860544 deg

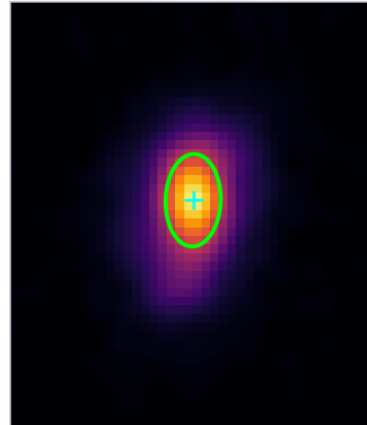
CEERS | physical source 00787

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.597

Eccentricity 0.802

Bar PA 128.7 deg

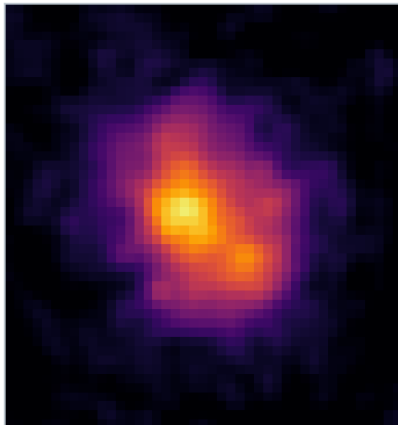
Estimated bar length _____

Galaxy sky center RA 214.987666 deg | Dec +52.870017 deg

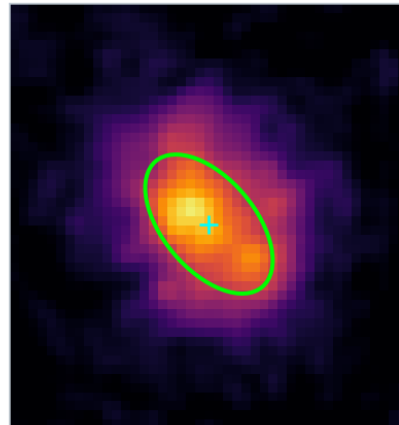
CEERS | physical source 00804

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.557

Eccentricity 0.830

Bar PA 170.8 deg

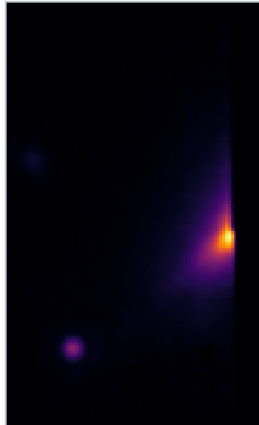
Estimated bar length _____

Galaxy sky center RA 215.036655 deg | Dec +52.905223 deg

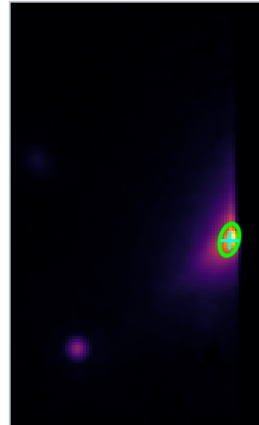
CEERS | physical source 00829

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.577

Eccentricity 0.817

Bar PA 115.9 deg

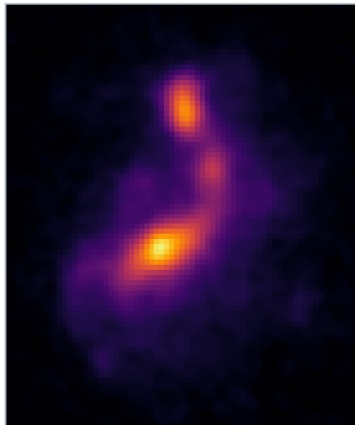
Estimated bar length _____

Galaxy sky center RA 214.967750 deg | Dec +52.857751 deg

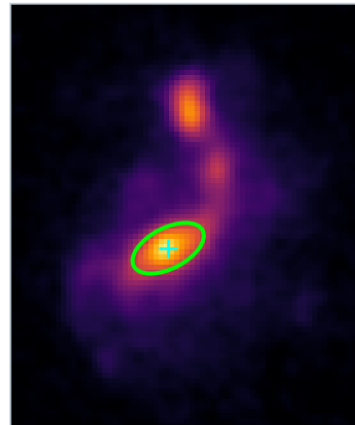
CEERS | physical source 01029

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.522

Eccentricity 0.853

Bar PA 67.9 deg

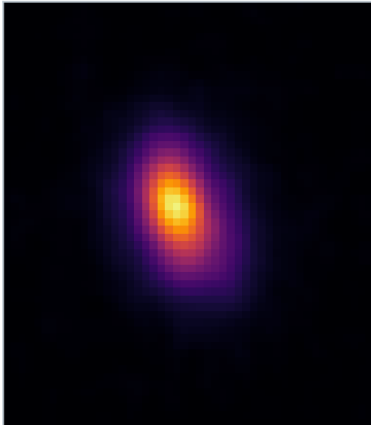
Estimated bar length _____

Galaxy sky center RA 215.044963 deg | Dec +52.919484 deg

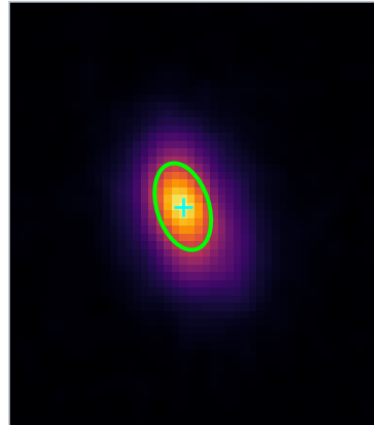
CEERS | physical source 01074

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.582

Eccentricity 0.813

Bar PA 148.3 deg

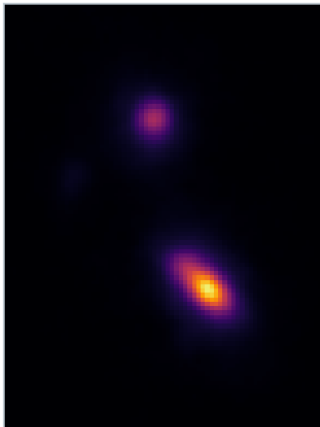
Estimated bar length _____

Galaxy sky center RA 215.019849 deg | Dec +52.903557 deg

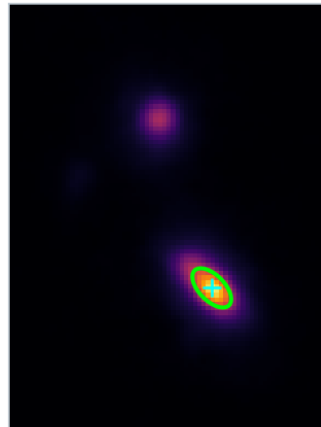
CEERS | physical source 01110

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.522

Eccentricity 0.853

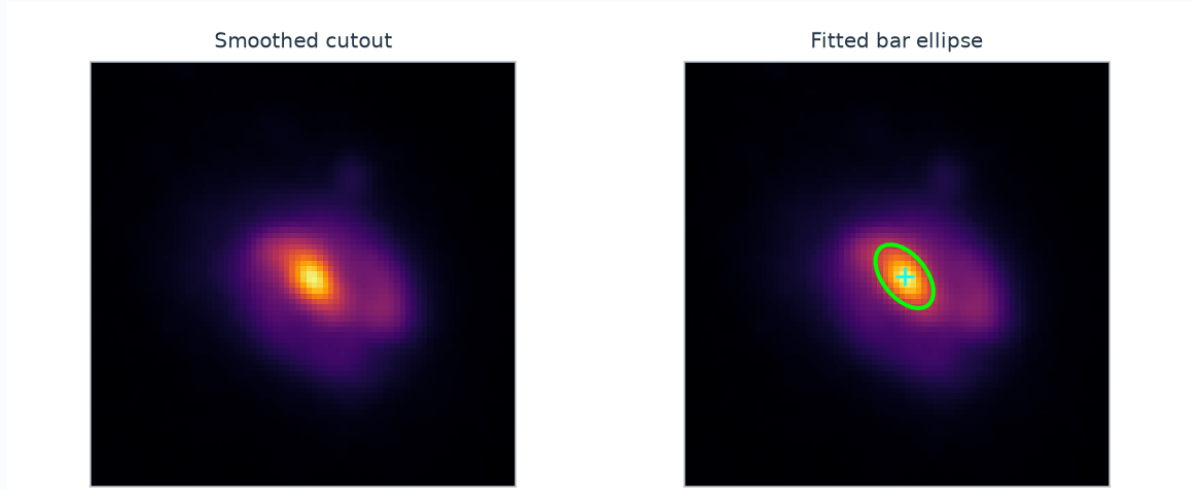
Bar PA 175.0 deg

Estimated bar length _____

Galaxy sky center RA 214.975855 deg | Dec +52.874014 deg

CEERS | physical source 01156

Weakly Barred



Bar b/a 0.585

Eccentricity 0.811

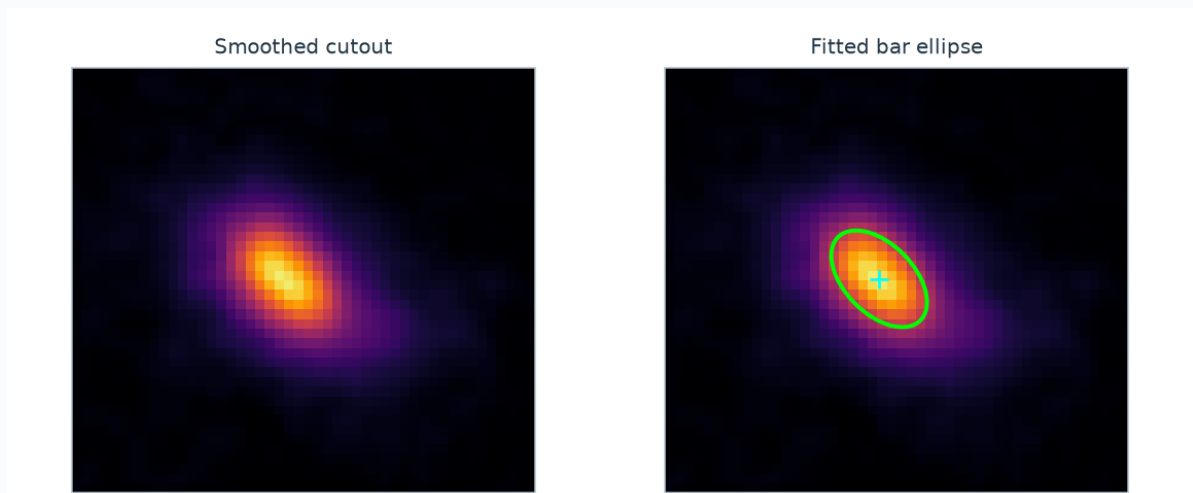
Bar PA 170.2 deg

Estimated bar length _____

Galaxy sky center RA 215.012864 deg | Dec +52.901425 deg

CEERS | physical source 01181

Weakly Barred



Bar b/a 0.590

Eccentricity 0.807

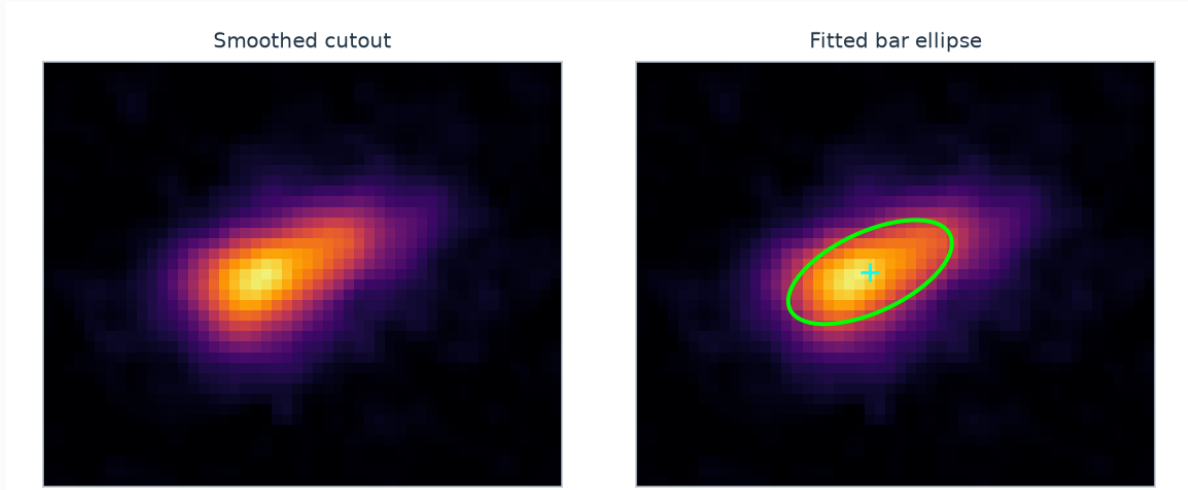
Bar PA 175.0 deg

Estimated bar length _____

Galaxy sky center RA 215.041748 deg | Dec +52.922430 deg

CEERS | physical source 01223

Weakly Barred



Bar b/a 0.452

Eccentricity 0.892

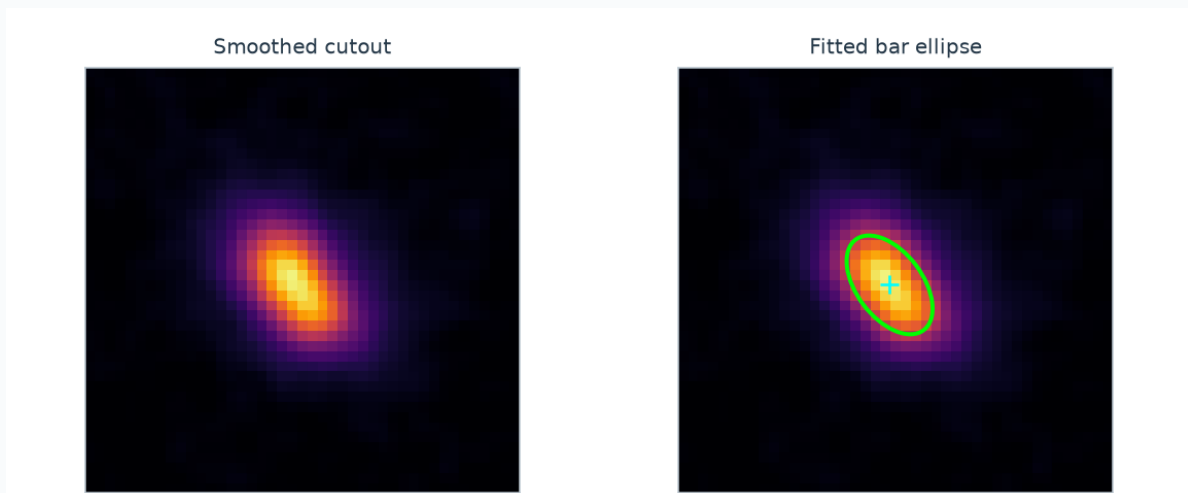
Bar PA 65.4 deg

Estimated bar length _____

Galaxy sky center RA 215.042979 deg | Dec +52.925085 deg

CEERS | physical source 01267

Weakly Barred



Bar b/a 0.581

Eccentricity 0.814

Bar PA 167.4 deg

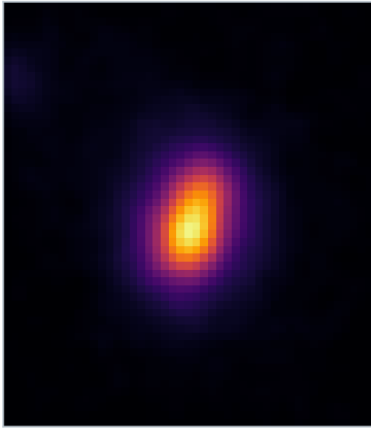
Estimated bar length _____

Galaxy sky center RA 214.967273 deg | Dec +52.873319 deg

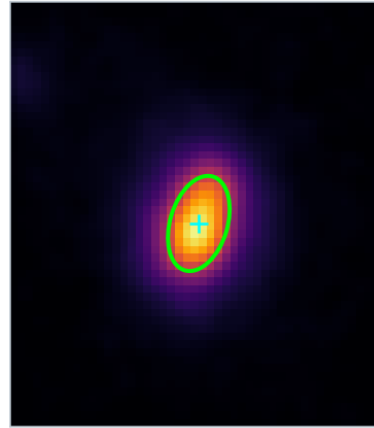
CEERS | physical source 01274

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.587

Eccentricity 0.810

Bar PA 113.6 deg

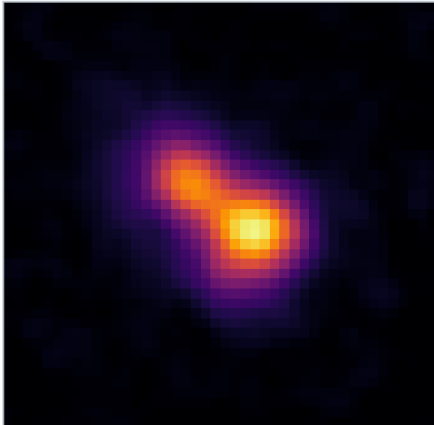
Estimated bar length _____

Galaxy sky center RA 214.962368 deg | Dec +52.870040 deg

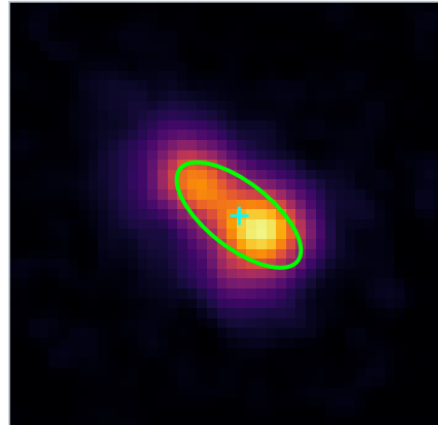
CEERS | physical source 01305

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.435

Eccentricity 0.900

Bar PA 2.4 deg

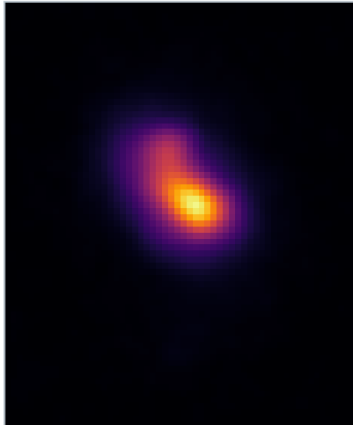
Estimated bar length _____

Galaxy sky center RA 214.988396 deg | Dec +52.889517 deg

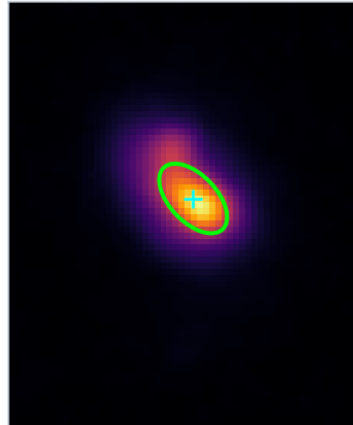
CEERS | physical source 01328

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.551

Eccentricity 0.835

Bar PA 174.1 deg

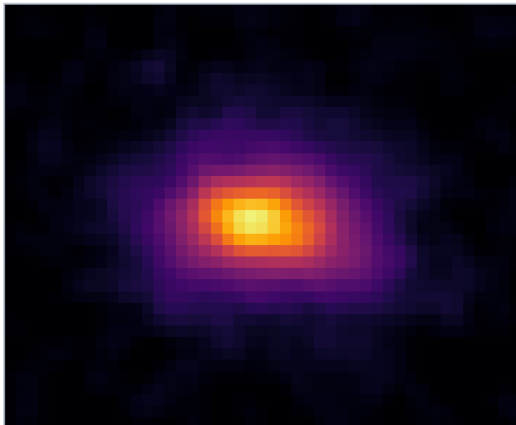
Estimated bar length _____

Galaxy sky center RA 214.954283 deg | Dec +52.866574 deg

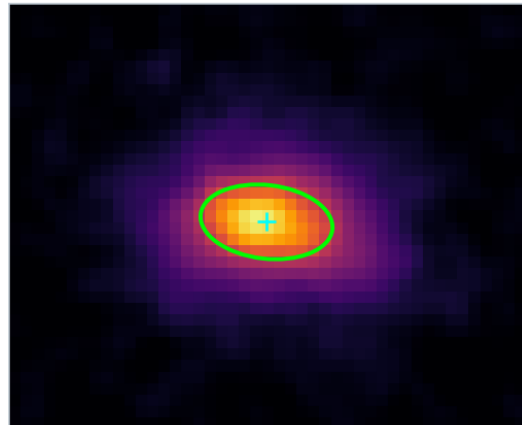
CEERS | physical source 01348

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.554

Eccentricity 0.832

Bar PA 33.8 deg

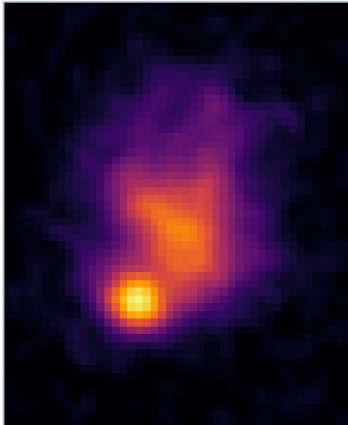
Estimated bar length _____

Galaxy sky center RA 215.038359 deg | Dec +52.926791 deg

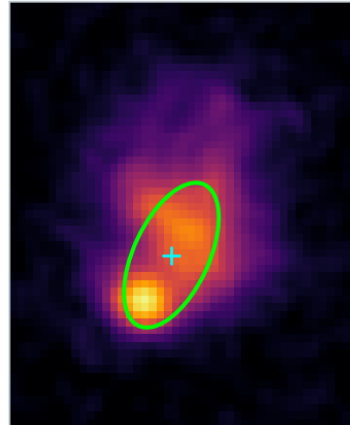
CEERS | physical source 01349

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.484

Eccentricity 0.875

Bar PA 105.4 deg

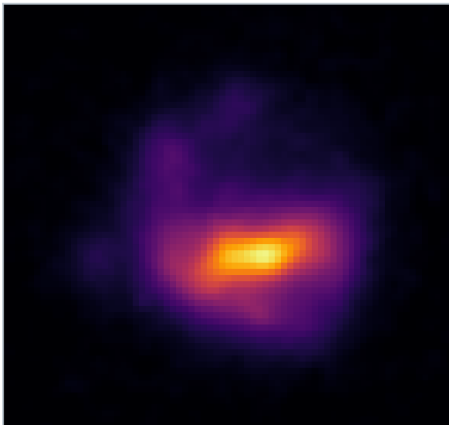
Estimated bar length _____

Galaxy sky center RA 215.023693 deg | Dec +52.916522 deg

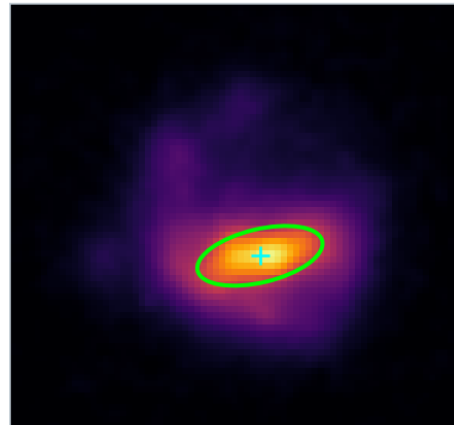
CEERS | physical source 01473

Weakly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.415

Eccentricity 0.910

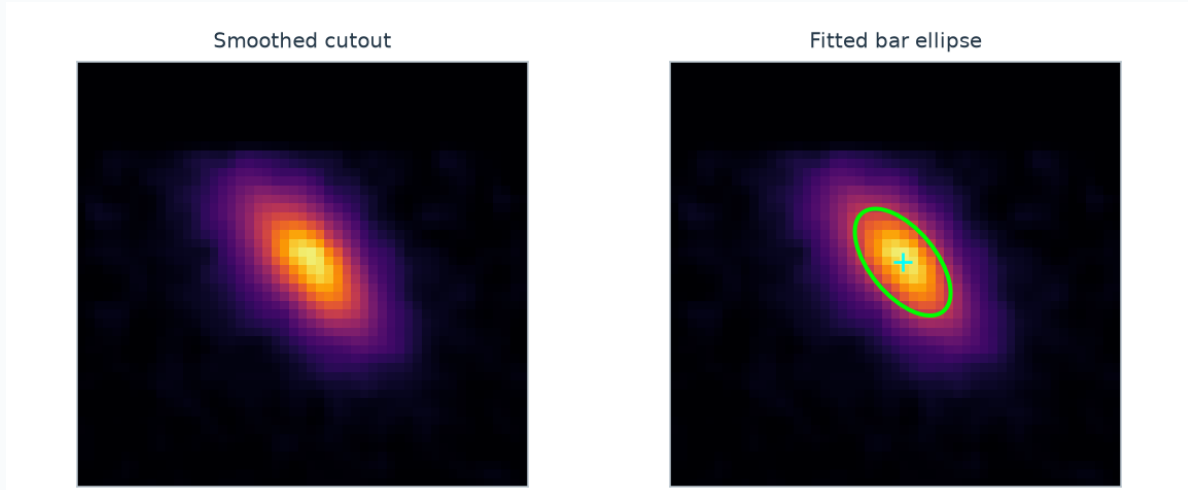
Bar PA 53.8 deg

Estimated bar length _____

Galaxy sky center RA 214.964226 deg | Dec +52.878649 deg

CEERS | physical source 01525

Weakly Barred



Bar b/a 0.518

Eccentricity 0.855

Bar PA 170.1 deg

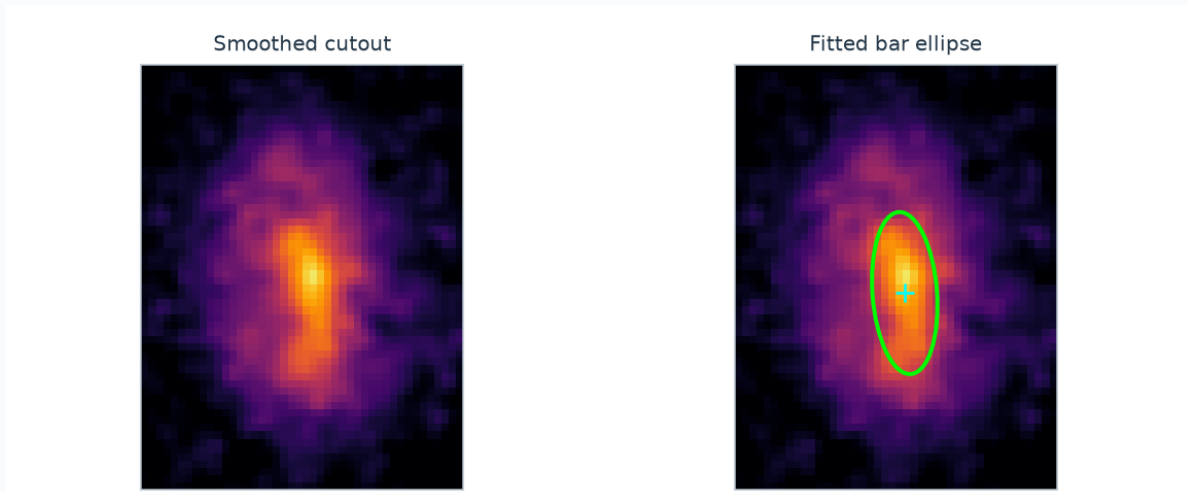
Estimated bar length _____

Galaxy sky center RA 215.029556 deg | Dec +52.926584 deg

Strongly Barred Galaxies

CEERS | physical source 00517

Strongly Barred



Bar b/a 0.398

Eccentricity 0.917

Bar PA 134.8 deg

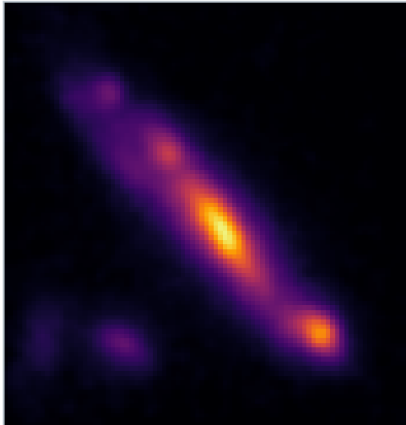
Estimated bar length _____

Galaxy sky center RA 215.045231 deg | Dec +52.900160 deg

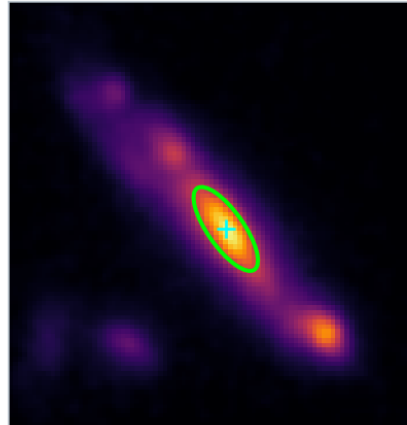
CEERS | physical source 00684

Strongly Barred

Smoothed cutout



Fitted bar ellipse



Bar b/a 0.360

Eccentricity 0.933

Bar PA 165.7 deg

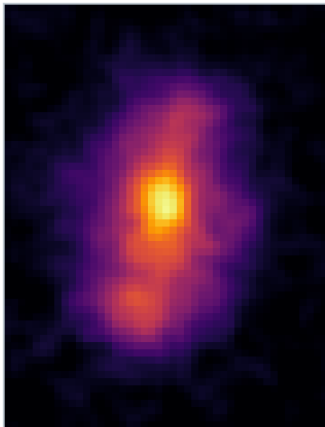
Estimated bar length _____

Galaxy sky center RA 215.027267 deg | Dec +52.894447 deg

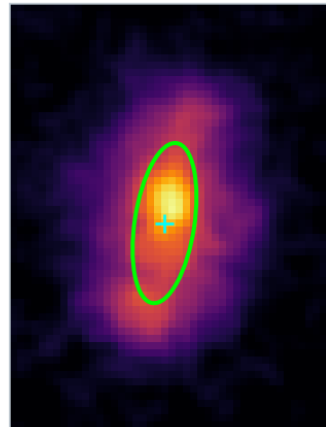
CEERS | physical source 01315

Strongly Barred

Smoothed cutout



Fitted bar ellipse



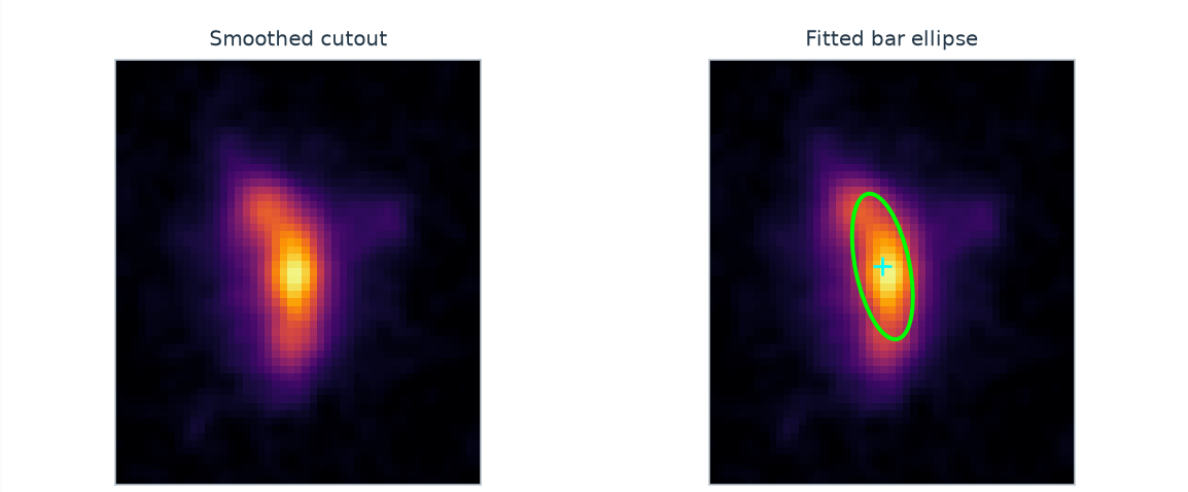
Bar b/a 0.370

Eccentricity 0.929

Bar PA 121.8 deg

Estimated bar length _____

Galaxy sky center RA 215.029836 deg | Dec +52.919177 deg



Bar b/a 0.366

Eccentricity 0.931

Bar PA 142.0 deg

Estimated bar length

Galaxy sky center RA 214.968972 deg | Dec +52.878859 deg